

CBSE Practice Worksheet

Class 7 | Mathematics | Difficulty: Hard | Set A

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. If $2\blacksquare = 3\blacksquare = 6\blacksquare$, which is true?

- (A) $1/z = 1/x + 1/y$
- (B) $z = x + y$
- (C) $z = xy$
- (D) $z = x - y$

2. Simplify: $(a^2 - b^2)/(a + b)$

- (A) $a - b$
- (B) $a + b$
- (C) $a^2 - b$
- (D) $a - b^2$

3. Find HCF of $(x^2 - 4)$ and $(x^2 - 4x + 4)$

- (A) $x - 2$
- (B) $x + 2$
- (C) $(x-2)^2$
- (D) $x^2 - 4$

4. Solve: $2^{(x+1)} = 32$

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$

5. Simplify: $(x^3 - 8)/(x - 2)$

- (A) $x^2 + 2x + 4$
- (B) $x^2 - 2x + 4$
- (C) $x^2 + 4$
- (D) $x^2 - 4$

6. If $\log\blacksquare 8 = x$, find x

- (A) 2
- (B) 3
- (C) 4
- (D) 8

7. If $x + 1/x = 5$, find $x^2 + 1/x^2$

- (A) 21
- (B) 23
- (C) 25
- (D) 27

8. What is the cube root of -125?

- (A) 5
- (B) -5
- (C) ± 5

(D) undefined

9. Find: $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$

(A) 2

(B) 8

(C) 15

(D) $\sqrt{15}$

10. Factorize: $x^2 - 5x - 14$

(A) $(x-7)(x+2)$

(B) $(x+7)(x-2)$

(C) $(x-7)(x-2)$

(D) $(x+7)(x+2)$

Section B: Fill in the Blanks (1 mark each)

11. If $a + b + c = 0$, then $a^3 + b^3 + c^3 = \underline{\hspace{2cm}}$

12. $(x + 1/x)^2 - (x - 1/x)^2 = \underline{\hspace{2cm}}$

13. If $x^2 - 6x + 8 = 0$, then $x = \underline{\hspace{2cm}}$ or $\underline{\hspace{2cm}}$

14. If $\log x = 2$, then $x = \underline{\hspace{2cm}}$

15. If $5^x = 125$, then $x = \underline{\hspace{2cm}}$

16. $\sqrt[3]{-64} = \underline{\hspace{2cm}}$

17. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - \underline{\hspace{2cm}}$

18. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2 = \underline{\hspace{2cm}}$

Section C: Short Answer Questions (2 marks each)

19. Prove: $(a + b)^3 - (a - b)^3 = 2b(3a^2 + b^2)$

20. Find the value of x : $3^{(2x-1)} = 27^{(x-2)}$

21. Solve: $|x - 3| + |x + 2| = 7$

22. Solve: $\log_{10}(x - 1) + \log_{10}(x + 1) = 3$

23. Solve: $x^2 - 7x + 12 = 0$

24. Simplify: $(2 + \sqrt{3})/(2 - \sqrt{3})$

25. Factorize: $8x^3 - 27$

--- End of Worksheet ---

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